

Minor CapStone Project Presentation (2026)

ExpenseEase: Reimbursement Portal

A Comprehensive Claim Management System

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Introduction

Web application for submitting and tracking expense reimbursement claims inside an organization.

Employees file claims, managers review their team's claims, admins manage users and oversee all claims.

Goals: faster turnaround, clear status (submitted / approved / rejected), role-based workflows, and centralized data in a database.

Built as a full-stack project with a REST API and a browser-based UI.

Tech Stack

HTML



CSS



{.js}

JavaScript

Layer	Technology
Backend	Java 17, Spring Boot 4.x
API	Spring Web MVC, REST controllers
Persistence	Spring Data JPA, Hibernate
Database	PostgreSQL
Validation	Jakarta Bean Validation (spring-boot-starter-validation)
Build	Maven
Frontend	Static HTML, CSS, JavaScript


Java


spring

PostgreSQL

Features and Functionality

Authentication

Login via POST
/auth/login
Frontend stores user
context (including
X-USER-ID
header).

Roles

ADMIN —
create/delete users, list all
users, assign managers to
employees, view/delete any
claim
MANAGER — review claims,
approve/reject with comments
EMPLOYEE — submit claims
view own claims

Claims lifecycle

Statuses: SUBMITTED,
APPROVED, REJECTED.
Submit, list claims,
pagination and optional
status filter,
get by id,
approve, reject,
admin delete.

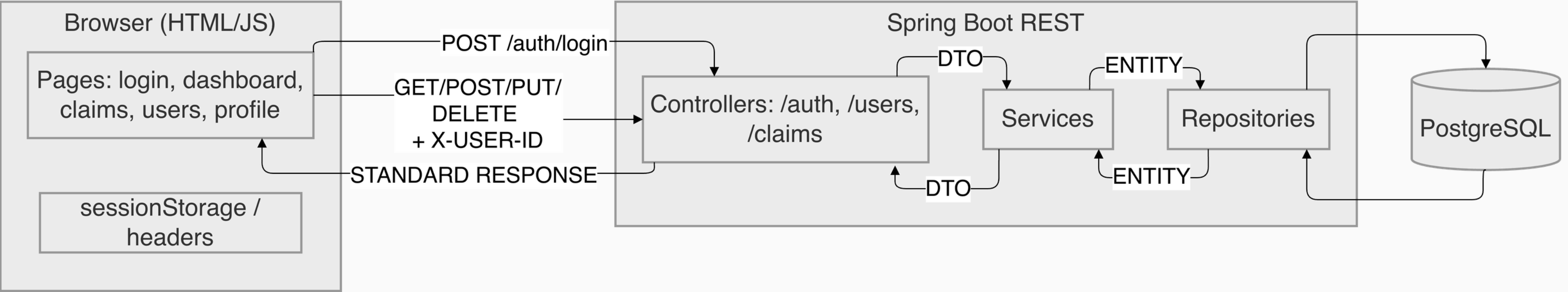
User management

CRUD-style operations:
create user,
list all,
filter by role,
assign manager,
delete user.

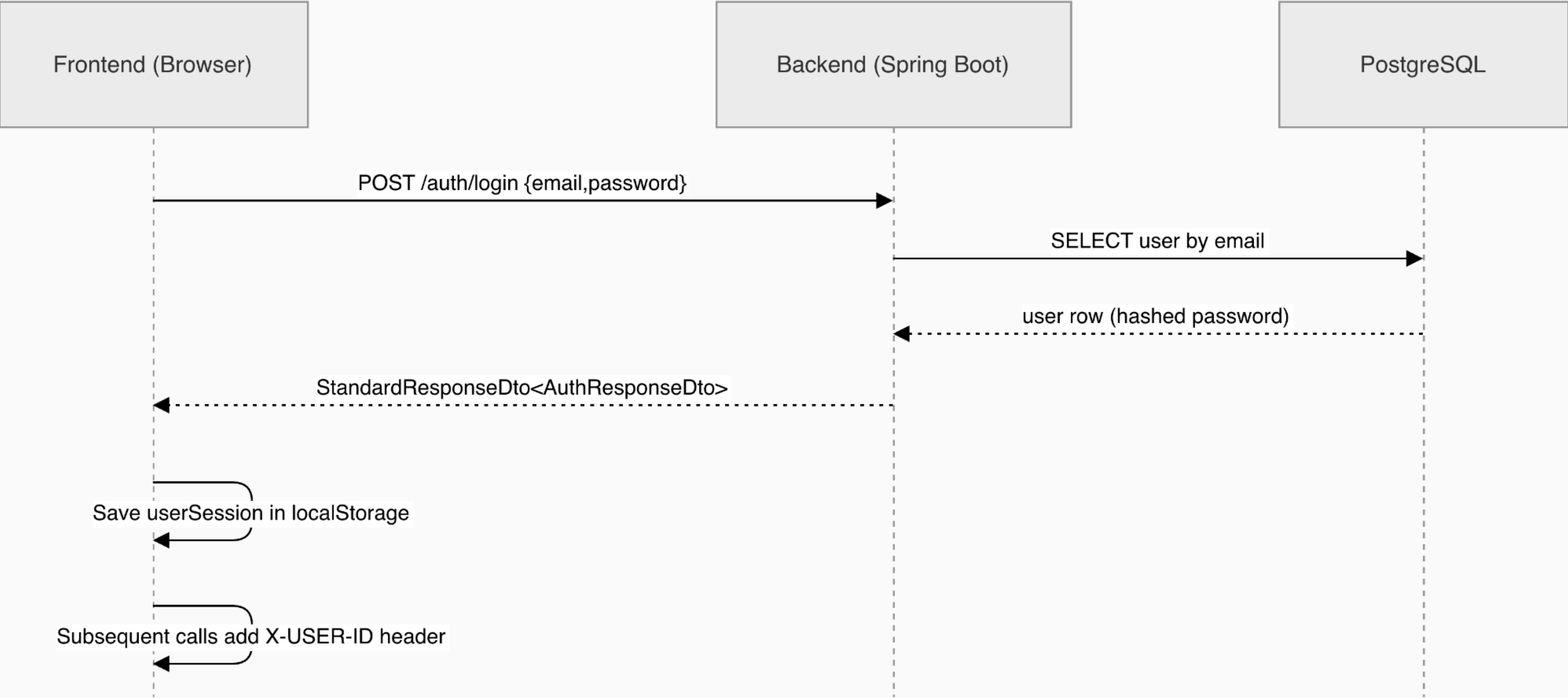
Quality / UX

Standard JSON
responses
(StandardResponseDto),
Global exception
handling, validation on
DTOs.

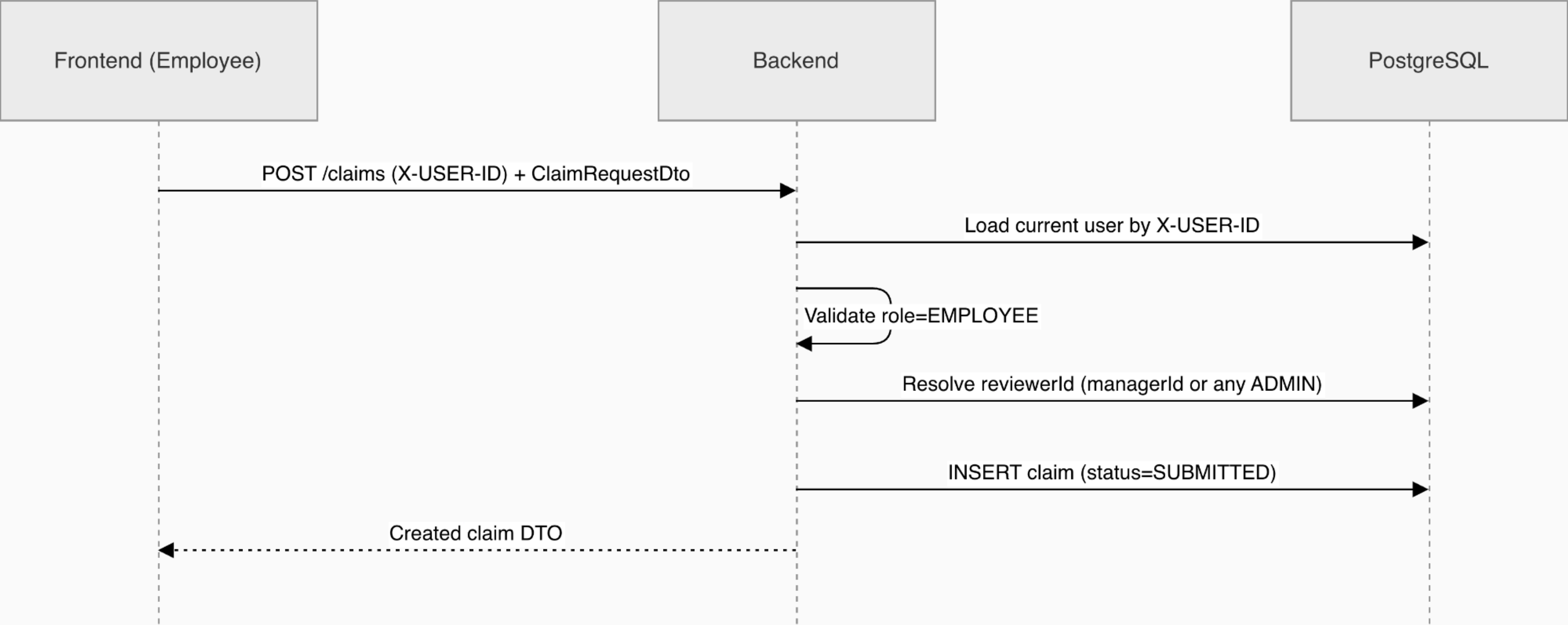
High-level request flow



Login workflow

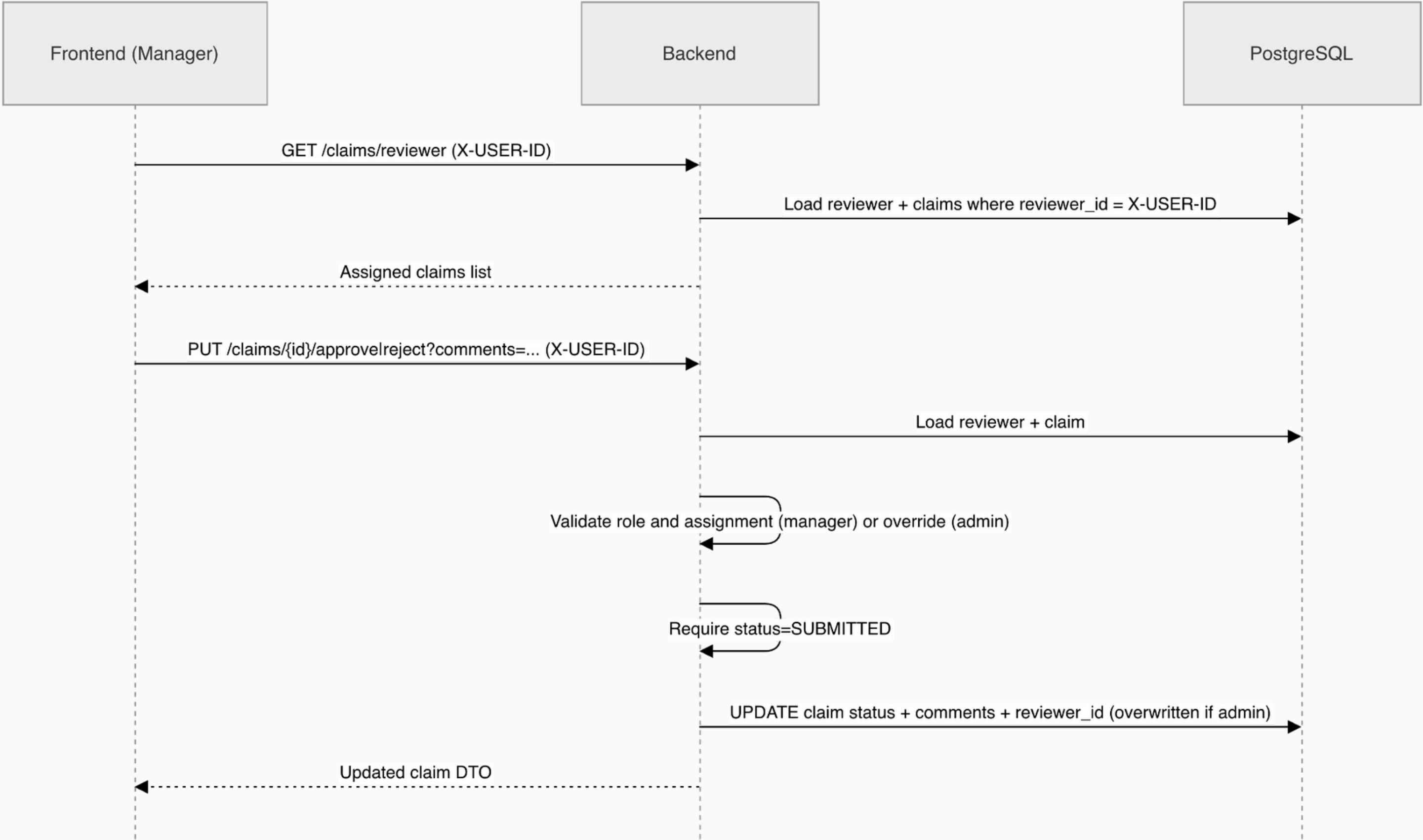


Claim submission workflow (employee)

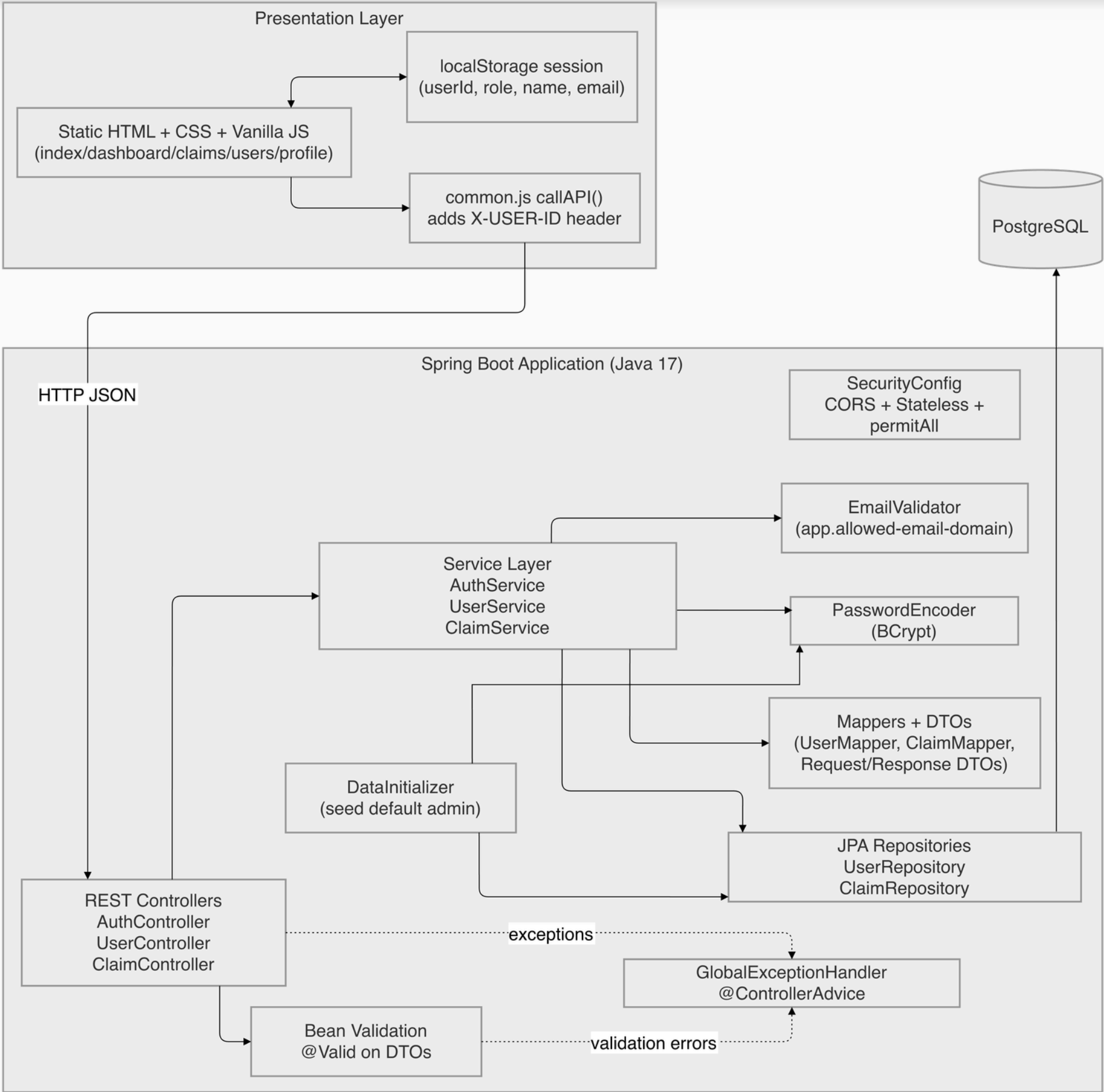


Data Flow Diagram

Review workflow (manager/admin)



Architecture Diagram



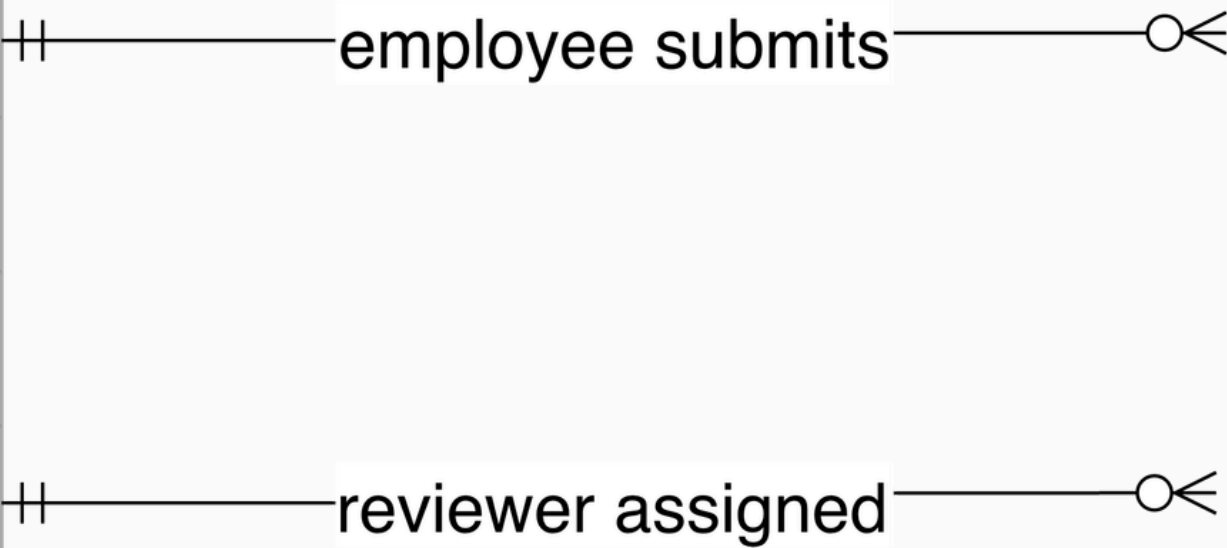
Entity-Relationship Diagram

users		
int	id	PK
string	name	
string	email	
string	password	
string	role	
int	manager_id	FK
datetime	created_at	

⊥

⊙

manager of



claims		
int	id	PK
float	amount	
date	claim_date	
string	title	
string	description	
string	status	
int	employee_id	FK
int	reviewer_id	FK
string	comments	
datetime	created_at	
datetime	updated_at	

ExpenseEase

Submit expense claims, track approvals, and get reimbursed — a simple flow for employees, managers, and admins.

Login

Email

Password

Password must be more than 6 characters (max 100).

Sign In

ExpenseEase

Reimbursement portal

Dashboard

Users

Claims

Nova Admin (ADMIN)

Profile

Logout

Dashboard

Welcome back, Nova Admin

Organization-wide view — all users and claims in the system.



USERS

20

Accounts you can administer



CLAIMS

20

Total expense claims



PENDING REVIEW

11

Awaiting a decision



APPROVED AMOUNT

₹23,331.10

Sum of approved claims

Recent claims

Latest activity from your scope

Office snacks misc

Claim #20 · ₹475.00 · 2025-11-07

SUBMITTED

Missing receipts

Claim #19 · ₹8,000.00 · 2025-07-29

REJECTED

Personal vacation mixed

Claim #18 · ₹15,000.00 · 2025-06-02

REJECTED

Future parking

Claim #17 · ₹300.00 · 2025-12-20

REJECTED

Over-budget workstation

Claim #16 · ₹22,000.00 · 2025-09-01

REJECTED

Client gift basket

APPROVED

ExpenseEase

Reimbursement portal

Dashboard

Users

Claims

Nova Admin (ADMIN)

Profile

Logout

Users

+ New User

Search by ID, name

All roles

All managers

ID	NAME	EMAIL	ROLE	MANAGER	ACTIONS
1	Nova Admin	admin@company.com	ADMIN	-	View Delete
2	Mira Gupta	mira.gupta@company.com	MANAGER	-	View Delete
3	Rohan Mehta	rohan.mehta@company.com	MANAGER	-	View Delete
4	Sonia Pillai	sonia.pillai@company.com	MANAGER	-	View Delete
5	Aarav Sharma	aarav.sharma@company.com	EMPLOYEE	Mira Gupta	View Delete
6	Isha Kapoor	isha.kapoor@company.com	EMPLOYEE	Mira Gupta	View Delete
7	Vikram Desai	vikram.desai@company.com	EMPLOYEE	Mira Gupta	View Delete
8	Neha Reddy	neha.reddy@company.com	EMPLOYEE	Mira Gupta	View Delete
9	Karan Joshi	karan.joshi@company.com	EMPLOYEE	Mira Gupta	View Delete
10	Priya Nair	priya.nair@company.com	EMPLOYEE	Mira Gupta	View Delete

ExpenseEase

Reimbursement portal

Dashboard

Users

Claims

Nova Admin (ADMIN)

Profile

Logout

Claims

All statuses

ID	EMPLOYEE	TITLE	AMOUNT	STATUS	REVIEWER	COMMENTS	ACTIONS
20	Lina Thomas	Office snacks misc	₹475.00	SUBMITTED	Sonia Pillai	-	View Review Delete
19	Ananya Bose	Missing receipts	₹8,000.00	REJECTED	Rohan Mehta	Upload invoices and GST.	View Delete
18	Dev Malik	Personal vacation mixed	₹15,000.00	REJECTED	Rohan Mehta	Separate business itinerary required.	View Delete
17	Isha Kapoor	Future parking	₹300.00	REJECTED	Mira Gupta	Prepaid items not reimbursed.	View Delete
16	Aarav Sharma	Over-budget workstation	₹22,000.00	REJECTED	Mira Gupta	Use IT procurement catalogue only.	View Delete

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ExpenseEase

Reimbursement portal

Dashboard

Users

Claims

Nova Admin (ADMIN)

Profile

Logout

Profile

Your account

These details are read-only. You can update your password using the form beside this card.

User ID: 1

Name: Nova Admin

Email: admin@company.com

Role: ADMIN

Member since: 5/4/2026

Change password

Your sign-in password is the only field you can change here.

Current password

New password

Confirm new password

Any characters allowed; more than 6 and up to 100 characters.

Change password

THANK YOU